Iván Sevillano-García

521 citations | h-index 4 | 512 since 2021

Researcher in Trustworthy & Explainable AI | PhD, Information & Communication Technologies, UGR 2025

isevillano@go.ugr.es | LinkedIn | GitHub | Scholar | ORCID | Granada, Spain

PROFILE

Researcher in AI with a focus on Explainable AI, OOD detection, and trustworthy deployment in high-stakes domains. I develop methods that improve interpretability, robustness, and reliability of deep learning systems, combining theoretical modeling with empirical evaluation on benchmark and real-world datasets. Contributions: frameworks for explanation auditing, explanation-driven regularization, and explainable OOD detection with statistical confidence. Published in Pattern Recognition, MIR, IEEE JBHI; speaker at ADIA Lab Summer School on XAI (2025). Currently advancing toward trustworthy agentic AI — harness engineering, loop engineering, persistent memory with SOTA LLMs.

SKILLS

- **Programming:** Python, R, PyTorch, TensorFlow, scikit-learn
- **HPC:** multi-GPU clusters (A100), Slurm, distributed training
- **Reproducibility:** Conda, TensorBoard, experiment tracking
- **DevOps:** Git, Linux, scientific computing
- **Peer review:** Information Sciences (2025), Information Fusion (2023)

RESEARCH AREAS

- Explainable AI (XAI)
- Trustworthy AI
- OOD detection

Education

PhD, Information & Communication Technologies 2021–2025
University of Granada — DaSCI Institute
Thesis: Explainable AI: Methods for the Generation, Evaluation, and Correction of Explanations
Supervisors: F. Herrera, J. Luengo Defense: Nov 13, 2025 **Sobresaliente Cum Laude**
Contributions: REVEL (explanation evaluation), X-SHIELD (explanation regularization), STOOD-X (explainable OOD detection).
hdl.handle.net/10481/112935 ISBN 9791370150433

M.Sc. Data Science & Computer Engineering 2019–2020
University of Granada

B.Sc. Double Degree in CS & Mathematics 2013–2019
University of Granada — ETSIIT

Research Experience

Predoctoral Researcher 2019–2025
University of Granada / DaSCI Institute
- Trustworthy AI methodologies: explanation evaluation, explanation-driven regularization, explainable OOD detection.
- First-author in Pattern Recognition, Machine Intelligence Research, Information Systems Frontiers, IJIS.
- Collaborated on COVID-19 chest X-ray diagnosis (IEEE JBHI, 477 cit.) and metallographic segmentation for the steel industry (Information Fusion, 59 cit.).
- Researcher within the ENIA IAFER Chair (EU Next Generation-funded program on responsible AI).

Visiting Researcher Feb–Mar 2026
Poznan University of Technology, Poland — host: Prof. J. Stefanowski
- Prototype-based and counterfactual explanations for XAI post-hoc methods; joint publication proposal in progress.

Industrial Collaboration — ArcelorMittal 2020–2021
via DaSCI, UGR
- Deep-learning ensemble for metallographic image segmentation; MetalDAM dataset (Information Fusion, 59 cit.).

Selected Publications

STOOD-X: Explainable OOD Detection via Nonparametric Statistical Testing on Large-Scale Datasets.
Pattern Recognition, 2026. I. Sevillano-García, J. Luengo, F. Herrera. DOI
X-SHIELD: Regularization for Explainable Artificial Intelligence.
Machine Intelligence Research, 23, 528–544, 2026. I. Sevillano-García, J. Luengo, F. Herrera. DOI
A Three-Level Framework for LLM-Enhanced eXplainable AI.
Information Systems Frontiers, 2025. M. Bello, R. Bello, M.M. García, A. Nowé, I. Sevillano-García, F. Herrera. DOI
Fine-Grained Prostate Cancer Tissue Classification via Expert-Annotated Dataset & Interpretable DL.
IEEE CAI 2026, 1264–1269. J. Ayensa-Jiménez, D. Navarro, I. Sevillano-García, et al.
REVEL Framework to Measure Local Linear Explanations for Black-Box Models.
Intl. J. Intelligent Systems, 2023. I. Sevillano-García, J. Luengo, F. Herrera. DOI (15 cit.)
Optimizing LIME Explanations Using REVEL Metrics.
HAIS, 304–313, 2023. I. Sevillano-Garcia, J. Luengo, F. Herrera.
Low-Impact Feature Reduction Regularization Term.
xAI (LBW), 135–139, 2023. I. Sevillano-García, J. Luengo, F. Herrera.
A Tutorial on Segmentation of Metallographic Images: MetalDAM Dataset, DL Ensemble.
Information Fusion, 2022. J. Luengo, R. Moreno, I. Sevillano-García, et al. (59 cit.)
Slicer: Feature Learning for Class Separability with LS-SVM Loss (COVID-19 X-ray).
HAIS, 305–315, 2021. D. Charte, I. Sevillano-García, et al.
COVIDGR Dataset & COVID-SDNet Methodology for Predicting COVID-19 (Chest X-ray).
IEEE JBHI, 24(12), 3595–3605, 2020. S. Tabik, A. Gómez-Ríos, J.L. Martín-Rodríguez, I. Sevillano-García, et al. DOI (477 cit.)

ADIA Lab Engagement

Speaker, ADIA Lab International Summer School on Explainable AI May 2025
Granada — ADIA Lab (Abu Dhabi) in collaboration with UGR/DaSCI
- Delivered **"Lab on xAI Tools"**: hands-on introduction to AIX360 (AI Explainability 360), covering LIME, SHAP, ProtoDash, counterfactual explainers, and proxy metrics for explanation-quality benchmarking.
- Co-organized working-team activities for participating students. Listed among confirmed speakers alongside ADIA Lab Advisory Board members.

Teaching & Mentoring

- PhD student co-supervision (ongoing): Machine Unlearning, XAI metrics, XAI for AI agents.
- Co-supervision of 2 PhD students at UCLV (Cuba), collab. with Prof. M. Bello's group.